Project Title

Explainable AI Framework for Loan Default Prediction Using Ensemble Machine Learning

Project Report submitted

For the partial fulfilment of grade for

the subject

Applied Machine Learning for Text Analysis (24ALT3101)

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ABSTRACT

Loan default prediction plays a crucial role in helping financial institutions minimize credit risk and make informed lending decisions. Traditional machine learning models often achieve high prediction accuracy but lack interpretability, making it difficult for stakeholders to understand the factors influencing the predictions. This project proposes an Explainable Artificial Intelligence (XAI) framework for loan default prediction using ensemble machine learning techniques. The proposed system utilizes **Random Forest** and **XGBoost** algorithms to accurately classify borrowers based on their likelihood of default. The model is trained and evaluated using publicly available datasets such as the **Lending Club Dataset** and **Home Credit Default Risk Dataset**, which contain customer financial information and loan-related attributes. To improve transparency and trust, explainability techniques including **SHAP (SHapley Additive Explanations)** and **LIME (Local Interpretable Model-agnostic Explanations)** are incorporated to identify the key features contributing to each prediction. Furthermore, relevant textual fields available in loan application data are preprocessed and analyzed using text analysis techniques to extract meaningful information that enhances predictive performance. The proposed framework aims to provide an accurate, interpretable, and reliable decision-support system for financial institutions, enabling fair credit evaluation while supporting regulatory compliance and responsible AI practices.

PROBLEM STATEMENT

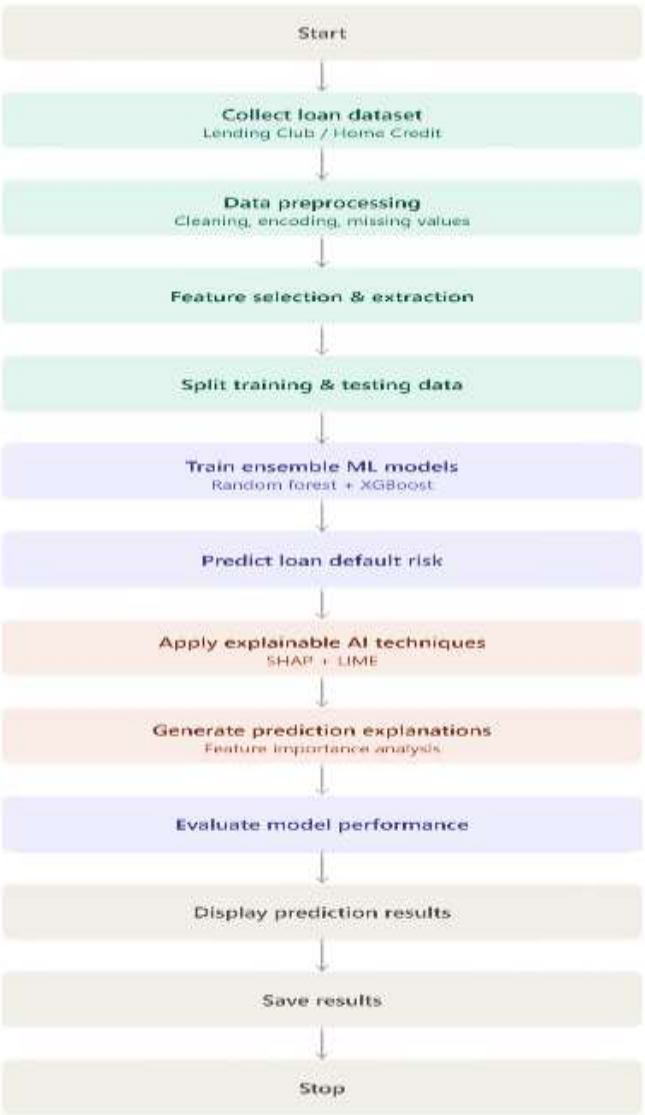
In banks and financial institutions, a large number of loan applications are processed every day, making it difficult to accurately identify customers who may default on their loans. Incorrect predictions can lead to financial losses, while rejecting eligible applicants may affect customer satisfaction and business growth. Traditional machine learning models can predict loan defaults, but they often fail to explain the reasons behind their decisions, reducing trust among users and financial analysts.

Due to the lack of transparency and interpretability, it becomes difficult to understand which factors influence the prediction or to detect possible bias in the decision-making process. Therefore, there is a need to develop a simple and effective Explainable AI framework that uses ensemble machine learning techniques to accurately predict loan defaults while providing clear explanations for each prediction, enabling more reliable and transparent credit risk assessment.

OBJECTIVES

1. To design and implement an Explainable AI-based loan default prediction framework using ensemble machine learning techniques.
2. To develop loan default prediction models using machine learning algorithms like XGBoost and Random Forest.
3. To apply Explainable AI techniques such as SHAP and LIME for interrupting model predictions and identifying important features.
4. To evaluate the performance of the proposed framework and provide transparent insights for credit risk assessment.

Flowchart (Project Structure):



CONCLUSION

In this project, an Explainable AI Framework for Loan Default Prediction Using Ensemble Machine Learning was successfully developed to predict the risk of loan defaults using financial data. The system uses ensemble learning algorithms such as **Random Forest and XGBoost** to improve prediction accuracy and identify potential high-risk borrowers. The Lending Club Dataset and Home Credit Default Risk Dataset are analyzed through data preprocessing and feature extraction techniques to build an effective prediction model.
Explainable AI techniques such as **SHAP and LIME** are integrated to provide insights into the factors influencing each prediction, making the model more transparent and trustworthy. This project provides a reliable and interpretable solution for credit risk assessment in financial applications. Overall, the project improves practical understanding of machine learning, text analysis, and explainable AI concepts while providing a foundation for developing advanced AI-based financial decision-support systems.

SOFTWARE REQUIREMENTS

1. **Operating System**: Windows 10/11 or Linux (Ubuntu)
2. **Programming Language**: Python 3.x
3. **Development Environment**: Jupyter Notebook / Google Colab / VS Code
4. **Libraries and Frameworks**:
    - Pandas – Data processing and analysis
    - NumPy – Numerical computations
    - Scikit-learn – Machine learning model development
    - XGBoost – Ensemble learning algorithm for prediction
    - SHAP – Explainable AI model interpretation
    - LIME – Local model explanation
    - Matplotlib / Seaborn – Data visualization
5. **Dataset**:
    - Lending Club Dataset
    - Home Credit Default Risk Dataset
6. **Database/Storage**: CSV files for dataset storage and result analysis